

Student Dropout Risk Prediction in Online Learning Platforms

1. Problem Statement

Imagine an online learning platform. Thousands of students enroll in courses, but only a small percentage actually finish them. Most students gradually stop participating and eventually drop out.

Instead of waiting until a student has already dropped out, we want to **predict early** which students are likely to leave the course.

The system will analyze students' learning behavior, such as:

- How often they log in
- Whether they watch lecture videos
- Whether they submit assignments on time
- How active they are in discussion forums
- Their assessment scores

Using this information, a Machine Learning model predicts whether each student is:

- **Low Risk**
- **Medium Risk**
- **High Risk** of dropping out.

If a student is predicted as high risk, instructors can intervene early by:

- Sending reminder emails
- Providing mentoring
- Offering counselling
- Giving additional learning resources

However, the prediction should also be fair. For example, the model should not unfairly classify older students or students with disabilities as high risk simply because of historical patterns in the data. Therefore, we also evaluate fairness metrics before deploying the model.

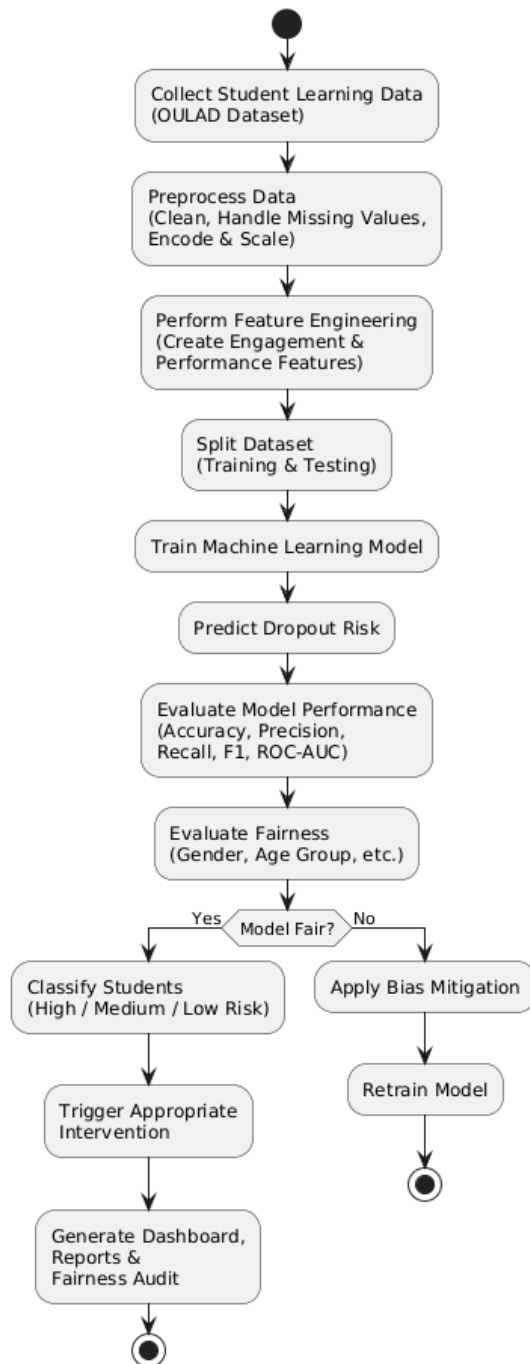
Summary of the Project

1. Collect student learning data from the OULAD dataset.
2. Clean and preprocess the data.

3. Create meaningful features that represent student engagement and performance.
4. Train a machine learning model to predict whether a student is likely to drop out.
5. Evaluate the model using metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.
6. Perform a fairness audit to ensure predictions are unbiased across demographic groups.
7. Categorize students into Low, Medium, and High dropout risk.
8. Trigger appropriate interventions (alerts, mentoring, counselling, or monitoring).
9. Provide dashboards and reports for instructors and administrators to support timely and ethical decision-making.

Flowchart – Student Dropout Risk Prediction System

Overall Workflow - Student Dropout Risk Prediction System



Explanation:

Imagine you are managing an online learning platform where thousands of students enroll in courses. Some students complete the course, while many stop participating after a few weeks. The goal of this project is to identify **students who are likely to drop out early** so that instructors can help them before they leave the course.

The entire project follows the workflow shown above.

Step 1: Collect Student Learning Data

The first step is to collect data from the **Open University Learning Analytics Dataset (OULAD)**.

This dataset contains information about students and their learning behaviour, such as:

- Student demographic information
- Number of logins
- Clickstream activity
- Video viewing records
- Assignment submissions
- Assessment scores
- Forum participation
- Final course outcome (Completed or Dropped Out)

The dataset becomes the input for the machine learning model.

Step 2: Preprocess the Data

Raw data usually contains problems such as:

- Missing values
- Duplicate records
- Incorrect data formats
- Categorical values that computers cannot understand

Therefore, the data is cleaned before training the model.

Typical preprocessing includes:

- Removing duplicate records
- Filling missing values
- Encoding categorical variables
- Normalizing numerical values
- Removing unnecessary columns

After preprocessing, the data becomes consistent and ready for analysis.

Step 3: Feature Engineering

The raw data does not always provide the best information for prediction.

Feature Engineering creates more meaningful variables from existing data.

Examples include:

- Average number of logins per week
- Percentage of completed videos
- Number of late assignment submissions
- Assessment average
- Forum activity score
- Student engagement score

These new features help the machine learning model recognize learning patterns more effectively.

Step 4: Split the Dataset

The dataset is divided into two parts:

Training Dataset

Used to teach the machine learning model.

Usually about **70–80%** of the data.

Testing Dataset

Used to evaluate how well the model performs on unseen students.

Usually about **20–30%** of the data.

This ensures that the model is not simply memorizing the training data.

Step 5: Train the Machine Learning Model

Now the training data is used to build a prediction model.

Possible algorithms include:

- Logistic Regression
- Random Forest
- XGBoost
- Decision Tree

The model learns patterns from historical student behaviour.

For example, it may learn that students who:

- rarely log in,
- miss assignments,
- and have low quiz scores

are more likely to drop out.

Step 6: Predict Student Dropout Risk

After training, the model predicts the likelihood that each student will drop out.

Instead of simply saying "Dropout" or "Not Dropout," it often generates a probability.

Example:

Student Dropout Probability

Student A 0.92

Student B 0.68

Student C 0.18

These probabilities help classify students into different risk levels.

Step 7: Evaluate Model Performance

Before using the model in a real educational platform, we need to check how accurate it is.

Common evaluation metrics include:

- **Accuracy** – Overall percentage of correct predictions.
- **Precision** – Among students predicted to drop out, how many actually did.

- **Recall** – Among all students who dropped out, how many were correctly identified.
- **F1 Score** – A balance between Precision and Recall.
- **ROC-AUC** – Measures how well the model distinguishes between students who will drop out and those who will not.

For this capstone, the target is an **AUC-ROC of at least 0.78**.

Step 8: Evaluate Fairness

A model can be accurate but still unfair.

For example, if historical data contains biases, the model might incorrectly predict higher dropout risk for certain demographic groups, such as:

- Older students
- Students with disabilities
- Students from underrepresented backgrounds

This does not necessarily reflect their actual learning behaviour.

Therefore, fairness is evaluated across demographic groups using fairness metrics such as:

- Demographic Parity
- Equal Opportunity
- Disparate Impact

If the model treats all groups fairly, it can move to the next stage.

If unfairness is detected, bias mitigation techniques are applied and the model is retrained.

This is why fairness is a key requirement of the project.

Step 9: Classify Students by Risk Level

Once the model is approved, each student is assigned a risk category.

High Risk

These students have a high probability of dropping out.

Example:

- Rarely log in
- Miss multiple assignments
- Low assessment scores

Medium Risk

These students show some warning signs but are still participating.

Low Risk

These students are actively engaged and are likely to complete the course successfully.

Step 10: Trigger Appropriate Interventions

Based on the predicted risk level, the system recommends suitable actions.

High Risk

- Immediate alert to instructor
- Personalized email reminders
- Academic counselling
- Mentor assignment

Medium Risk

- Follow-up messages
- Learning support
- Additional practice materials

Low Risk

- Continue regular monitoring

The goal is to help students before they disengage completely.

Step 11: Generate Dashboard and Reports

Finally, the system generates reports for instructors and administrators.

These reports include:

- Students at high risk

- Overall dropout statistics
- Model performance metrics
- Fairness audit results
- Suggested interventions
- Data governance and compliance summary

These reports help institutions make informed decisions and provide timely support to students.

Final Outcome of the Project

By following this workflow, the system can:

- Predict which students are likely to drop out early.
- Achieve a strong prediction performance (target **AUC-ROC ≥ 0.78**).
- Ensure predictions are fair across different demographic groups.
- Recommend timely interventions to improve student retention.
- Generate dashboards and governance reports for instructors and academic administrators.

In simple terms, **the project uses students' online learning activities to predict who may leave a course early, checks that the predictions are fair for everyone, and helps educators intervene in time to improve course completion rates.**